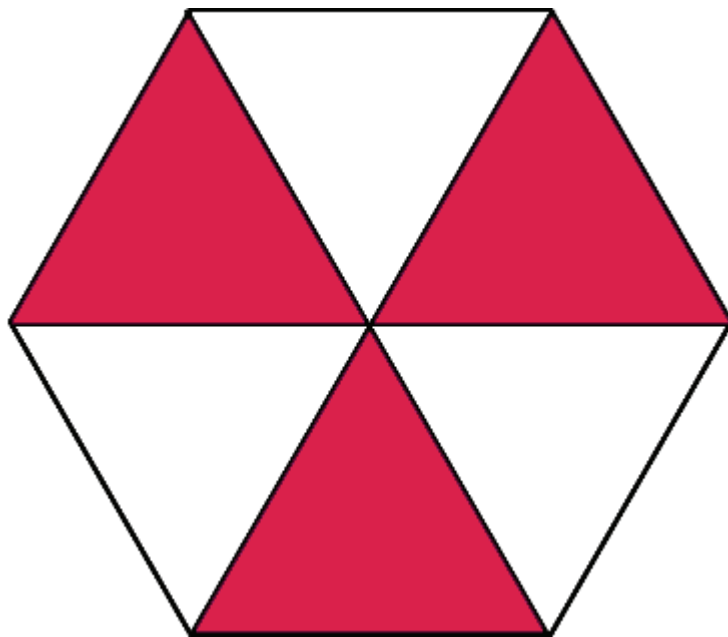


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Mathematical Modelling:

Mathematical Model

of an Epidemic



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Abstract

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Abstract

A mathematical model for simulating the spread of an epidemic through a susceptible population is implemented analytically via a system of differential equations and algorithmically using Python 3.5, with numerical solutions involving Euler's method. Simulation results based on the SIR model indicate that temporal variations in the susceptible $S(t)$, infected $I(t)$, and recovered $R(t)$ populations are dependent on the recovery and transmission parameters, α and β , respectively. Analytical expressions for $I(t)$ and $R(t)$ are derived and results verified graphically using Python. Further extensions to the original SIR model to include the effects of mortality, attenuation and super-spreader effects are proposed and the relevant simulations carried out in order to identify quantitative as well as qualitative trends.

List of variables

$S(t)$	susceptible population [persons]
$I(t)$	infected population [persons]
$R(t)$	recovered population [persons]
t	time [days]
α	recovery parameter [day^{-1}]
β	transmission rate parameter [$\text{persons}^{-1} \text{ day}^{-1}$]
μ	mortality parameter [day^{-1}]
$S(0)$	initial susceptible population [persons]
$I(0)$	initial infected population [persons]
$R(0)$	initial recovered population [persons]
β_0	initial transmission rate for time-varying linear attenuation [$\text{persons}^{-1} \text{ day}^{-1}$]
$I_{\max}$	maximum number of infected persons [persons]
$R_{\max}$	maximum number of recovered persons [persons]
$t_{\max}$	time taken to reach maximum infection [days]
$t_{\min}$	minimum time for full attenuation of disease [days]
Δt	step-size for Euler's method [days]

81.0 Introduction

An epidemic, according to the leading national public health institute in the United States, the Centers for Disease Control and Prevention (CDC), refers to “an increase, often sudden, in the number of cases of a disease above what is normally expected in that population in that area” [1]. When an epidemic is confined to localised geographic areas, it is more commonly known as an “outbreak”. In addition, an epidemiological definition of a pandemic has been cited in the 2011 Bulletin of the World Health Organisation (WHO) to be “an epidemic occurring worldwide, or over a very wide area, crossing international boundaries and usually affecting a large number of people” [2]. A limitation of this classical definition is that factors such as severity of the disease, population immunity and virology are not taken into consideration.

Based on the classical definition, although transboundary seasonal epidemics that occur annually in both hemispheres can cross international boundaries and affect large numbers of people, such seasonal epidemics are generally not classified as pandemics [2].

In recent history, infectious disease outbreaks that have reached epidemic proportions include the following [3]:

- Severe Acute Respiratory Syndrome (SARS), 2002 – 2003
- Middle East Respiratory Syndrome (MERS-CoV), 2012 – Present
- Ebola, 2013 – 2016
- Zika, 2015 – Present

SARS is the acronym for Severe Acute Respiratory Syndrome and is believed to have been transmitted via zoonotic pathways from civet cats to humans. The disease is as infectious as the common cold and according to [4,5], it was carried into Hong Kong by a medical doctor, whose overnight stay at a hotel in Kowloon resulted in several other guests being infected. This “super-spreader” event subsequently resulted in the disease being transmitted via air travel to other countries such as Canada, Singapore, Vietnam etc.

A distant relative of SARS, Middle East Respiratory Syndrome is caused by the MERS coronavirus and is believed to be transmitted from camels to humans, although the exact mechanism is still unclear. In June 2015, MERS had infected 50 patients in South Korea and the infection was linked to super-spreading events, viz. the index patient (Patient 01) had travelled to Saudi Arabia, Qatar, the United Arab Emirates and Bahrain before returning to South Korea, as well as Patient 14 who had transmitted the disease to other patients along with healthcare workers in a hospital emergency room. The latter was subsequently linked to nearly half of the MERS-infected patients in the country [6–8].

Back in February 2016, primarily due to clusters of microcephaly and other neurological disorders in Brazil and French Polynesia, Zika was declared by the WHO as a Public Health Emergency of International Concern (PHEIC) [9]. The Zika virus is transmitted by *Aedes* mosquitoes, the same insects that spread dengue, which are typically found in climates where it is not too cold for them to survive and are active even during daytime [10]. Infections with Zika have been linked to microcephaly, or babies being born with underdeveloped brains and below-average head size.

It is worth noting that historically, even though smallpox is one of the most contagious and virulent diseases known, it was declared to be eradicated in 1980 [11], after a worldwide immunisation programme spearheaded by the WHO.

82.0 Purpose and Motivation

Based on the discussion in preceding section, for the cases of SARS and MERS, the reasons for the emergent of super-spreaders is still not fully understood, particularly, whether these patients carry an exceptionally infectious strain of the virus, or if some other behavioural factors play a contributory role, e.g. proximity of the index patient to other individuals [10].

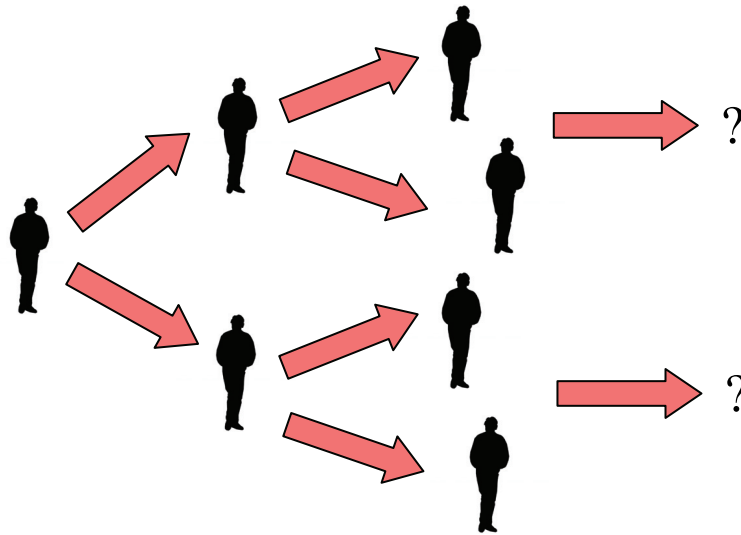


Figure 2.1: Schematic diagram representing spread of epidemic.

With these motivations in mind, the purpose of this mathematical modelling project is to numerically simulate the spread of an epidemic (Fig. 2.1) based on a system of differential equations and other associated parameters. The differential equations will be implemented and solved computationally in the form of simulations using Python 3.5. Based on the relevant graphical representations, quantitative as well as qualitative relationships between susceptible, infected and recovered populations will be examined.

83.0 Mathematical Analysis

3.1 System of Differential Equations: SIR Model

The proposed mathematical model for simulating the mechanism of the spread of an infectious disease (i.e. an epidemic) is based on a system of first-order differential equations, viz.

$$\frac{dS(t)}{dt} = -\beta S(t)I(t) \quad (\text{Eq. 3.1})$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \alpha I(t) \quad (\text{Eq. 3.2})$$

$$\frac{dR(t)}{dt} = \alpha I(t) \quad (\text{Eq. 3.3})$$

where $S(t)$, $I(t)$, $R(t)$ refer to the number of susceptible, infected and recovered persons in a given population, respectively. Collectively, Eqs. 3.1–3.3 are known as the SIR model [13]. α and β are constants of proportionality whose exact nature will be determined analytically in a subsequent section, as well as via computer simulations using Python.

With reference to Eq. 3.1, the rate of change of the susceptible population with time, $\frac{dS(t)}{dt}$, is proportional to the product of $S(t) \times I(t)$ because the expression can be understood intuitively based on the empirical fact that transmission of an infectious disease from one person to another takes place via *proximal contact*. Therefore, in this differential equation, direct contact is expressed in terms of the product between the susceptible population, $S(t)$, and infected population, $I(t)$.

Next, for Eq. 3.3, the time rate of change of recovered persons, $\frac{dR(t)}{dt}$, is necessarily a fraction of the number of infected persons, $I(t)$, where the constant of proportionality, α , determines the ratio of infected persons who are able to recover from the disease after infection and subsequently become immune.

Last but not least, for Eq. 3.2, the rate of change of infected persons with time, $\frac{dI(t)}{dt}$, is dependent on $\frac{dR(t)}{dt}$ subtracted from a product of three terms, since it is implicitly assumed in the SIR model that recovered persons become immune to the disease.

It is instructive to note at this stage that the following is an algebraic consequence of the model, i.e.

$$\frac{dS(t)}{dt} + \frac{dI(t)}{dt} + \frac{dR(t)}{dt} = 0 \quad (\text{Eq. 3.4})$$

In the next sub-section, dimensional analysis will be used to determine the units for constants of proportionality, α and β .

3.2 Dimensional Analysis

From Eq. 3.1,

$$\begin{aligned} \left| \frac{dS(t)}{dt} \right| &= \beta S(t) I(t) \\ \Rightarrow \beta &= \frac{\left| \frac{dS(t)}{dt} \right|}{S(t) \times I(t)} \end{aligned} \quad (\text{Eq. 3.5})$$

$$\Rightarrow [\beta] = \left[\frac{\frac{dS(t)}{dt}}{S(t) \times I(t)} \right] = \frac{\text{persons day}^{-1}}{\text{persons}^2}$$

$$\Rightarrow [\beta] = \text{persons}^{-1} \text{ day}^{-1} \quad (\text{Eq. 3.6})$$

Likewise, from Eq. 3.3,

$$[\alpha] = \left[\frac{\frac{dR(t)}{dt}}{I(t)} \right] = \frac{\text{persons day}^{-1}}{\text{persons}} = \text{day}^{-1} \quad (\text{Eq. 3.7})$$

3.3 Special Case of $\beta = 0$

At this stage, it is instructive to note that a special case exists where $\beta = 0$, which can be solved analytically. When $\beta = 0$, Eq. 3.2 becomes:

$$\frac{dI(t)}{dt} = -\alpha I(t) \quad (\text{Eq. 3.8})$$

which is a first-order linear differential equation that can be solved using the technique of separation of variables, i.e.

$$\int_{I_0}^I \frac{1}{I} dI = \int_0^t -\alpha dt$$

$$\Rightarrow \ln \left(\frac{I}{I_0} \right) = -\alpha t$$

$$\Rightarrow I = I_0 \exp(-\alpha t) \quad (\text{Eq. 3.9})$$

Eq. 3.9 implies that when $\beta = 0$, the number of infected persons should decrease exponentially with time. Consequently, the fact that β can be understood intuitively as the “transmission rate of the disease” will be analysed in more detail in Section 4.3.

3.4 Analytical Solution for Recovered Population $R(t)$

Based on Eq. 3.9, an analytical solution for the recovered population, $R(t)$, can be derived by substitution into Eq. 3.3:

$$\begin{aligned}\frac{dR(t)}{dt} &= \alpha I_0 \exp(-\alpha t) \\ \Rightarrow R(t) &= \alpha I_0 \times \int_0^t \exp(-\alpha t) dt \\ &= \alpha I_0 \times \left(-\frac{1}{\alpha}\right) [\exp(-\alpha t)]_0^t \\ \Rightarrow R(t) &= I_0 \{1 - \exp(-\alpha t)\} \quad (\text{Eq. 3.10})\end{aligned}$$

A corollary of Eq. 3.10 is that when $t = 0$, $R = 0$, indicating that as expected, the initial number of recovered persons is zero. Furthermore, as $t \rightarrow \infty$, $R \rightarrow I_0$, which implies that the initial infected population should recover fully, unless mortality is taken into consideration.

The effects of mortality on $R(t)$ will be examined via Python simulations in Section 5.1.

3.5 Euler's Method for Numerically Solving Differential Equations

In the subsequent Python simulations, the differential equations in the SIR model are solved numerically based on Euler's method, which is briefly outlined in this section.

For an initial value problem given by:

$$y' = f(t, y), \quad y(t_0) = y_0, \quad (\text{Eq. 3.11})$$

Euler's method [14] provides an approximation to the solution curve between $t = t_0$ and $t = t_0 + (n+1)\Delta t$, by a sequence of line segments joining the successive points (t_0, y_0) , (t_1, y_1) , $\dots$, (t_n, y_n) , (t_{n+1}, y_{n+1}) , where Δt is known as the step-size.

For every $0 \leq i \leq n$, the following calculations are performed iteratively by computer:

$$t_{i+1} = t_i + \Delta t \quad (\text{Eq. 3.12})$$

$$y_{i+1} = y_i + \Delta t \times f(t_i, y_i) \quad (\text{Eq. 3.13})$$

84.0 Computer Simulations Using Python 3.5

4.1 Special Case where $\beta = 0$ and $\alpha = 0.3$

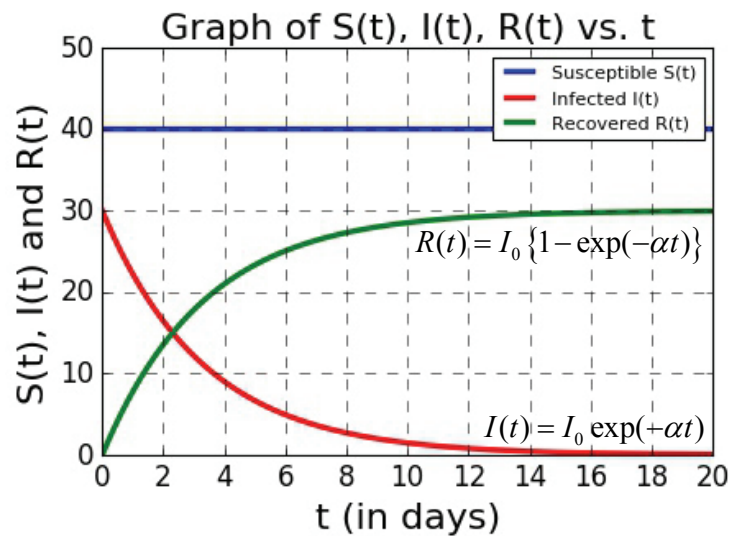


Figure 4.1: Exponential decrease in infected population for $\beta = 0$.

Initial Susceptible Persons $S(0)$	Initial Infected Persons $I(0)$	Initial Recovered Persons $R(0)$
40	30	0

Table 4.1: Initial conditions for simulation, assuming $\beta = 0$ and $\alpha = 0.3$.

The system of differential equations in Eqs. 3.1–3.3 were solved algorithmically using Python 3.5 and numerical solutions obtained via Euler’s method (see Appendix I to IV). When the initial infected population is set to $I(0) = 30$ and $\beta = 0$ (Table 4.1), the number of infected persons decrease exponentially with time, as observed in the simulation results of Fig. 4.1.

These numerical results are in agreement with Eq. 3.9, i.e. $I = I_0 \exp(-\alpha t)$, where the exponential decay term is based on the value $\alpha = 0.3$. Additionally, it can be implicitly deduced that constant β is the transmission rate term, since the number of susceptible persons remains constant (i.e. uninfected and healthy) when $\beta = 0$.

Furthermore, simulation results for the recovered population are fully consistent with the derived equation $R(t) = I_0 \{1 - \exp(-\alpha t)\}$ (Eq. 3.10), since initially $R(0) = 0$ but as time increases, $R(t)$ tends towards the initial number of infected persons, I_0 .

For clarity, both Eqs. 3.9 and 3.10 are shown on Figure 4.1.

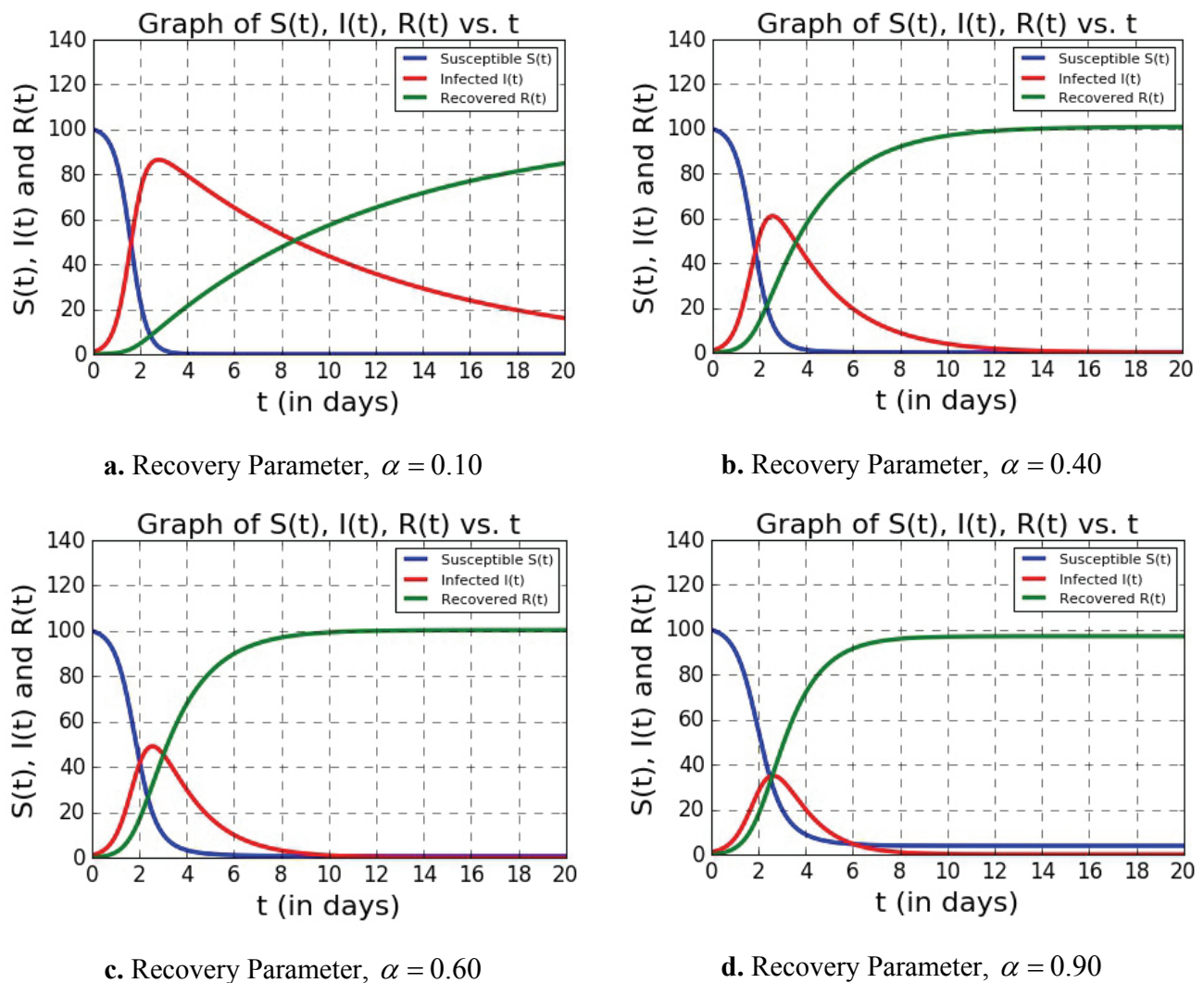
4.2 Variation of Recovery Parameter α with $\beta = 0.03$


Figure 4.2: Simulation graphs for variation of parameter α , with $\beta = 0.03$.

α (day^{-1})	Max. Infected Persons $I_{\max}$ (persons)	Time to Max. Infection $t_{\max}$ (days)
0.10	87	2.78
0.40	61	2.56
0.60	49	2.56
0.90	35	2.63

Table 4.2: Numerical results for $I_{\max}$ and $t_{\max}$ with variation of α .

Without any loss of generality, the transmission rate is set to be $\beta = 0.03$ and parameter α varied from 0.10 to 0.90 day^{-1} . From the simulation results (Fig. 4.2), it is observed that as α increases, the maximum number of infected persons decreases (Table 4.2). This observation could be understood in terms of the fact α represents the fraction of infected persons who recover from the disease, after being infected (i.e. the recovery parameter). In other words, the stronger the immunity of a given population, the fewer the maximum number of persons will become infected by a specific disease.

4.3 Variation of Transmission Parameter β with $\alpha = 0.3$

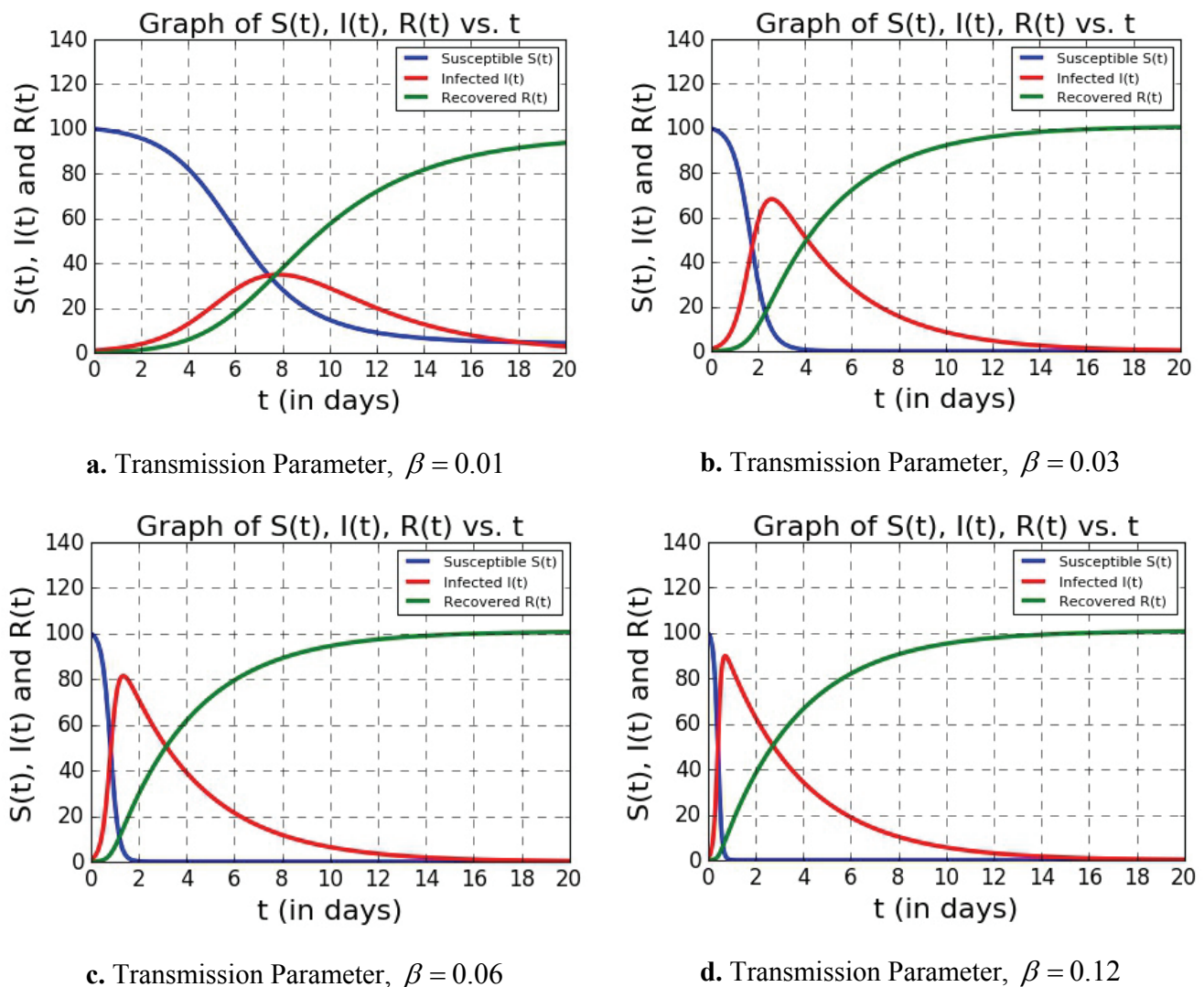
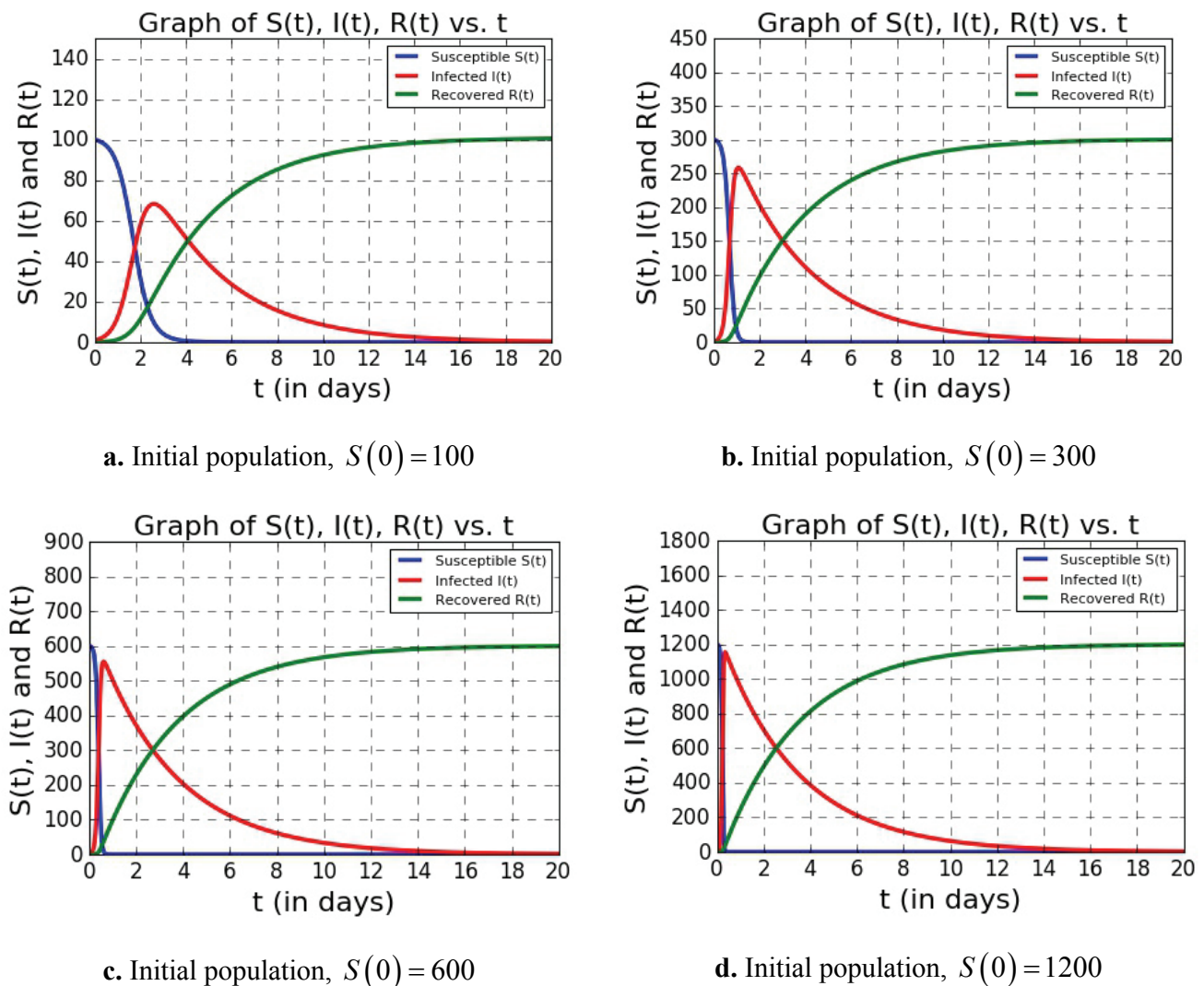


Figure 4.3: Simulation graphs for variation of parameter β , with $\alpha = 0.3$.

β (persons ⁻¹ day ⁻¹)	Max. Infected Persons $I_{\max}$ (persons)	Time to Max. Infection $t_{\max}$ (days)
0.01	35	7.83
0.03	69	2.58
0.06	82	1.36
0.12	90	0.72

Table 4.3: Numerical results for $I_{\max}$ and $t_{\max}$ with variation of β .

Next, in order to understand the physical significance of β , the parameter is varied incrementally from 0.01 to 0.12 persons⁻¹ day⁻¹. These values are chosen based on assertions from [15] that typically, $\beta < 1.00$. From the graphs in Fig. 4.3, the interpretation of β as the transmission rate term is clear, i.e. when β increases, the maximum number of infected persons increases rapidly, and the time taken to reach the point of maximum infection decreases (Table 4.3). Evidently, β is greater for a disease that is more contagious.

4.4 Variation of Initial Population $S(0)$ with $\alpha = 0.3$, $\beta = 0.03$

 Figure 4.4: Simulation graphs for variation of initial population $S(0)$.

$S(0)$	Max. Infected Persons $I_{\max}$ (persons)	Time to Max. Infection $t_{\max}$ (days)	Ratio $I_{\max} / S(0)$
100	69	2.58	0.69
300	259	1.08	0.86
600	555	0.63	0.93
1200	1155	0.36	0.96

 Table 4.4: Numerical results for $I_{\max}$, $t_{\max}$, and ratio $I_{\max} / S(0)$.

In this sub-section, the simulations results based on the SIR model are examined for different initial population sizes, i.e. $S(0) = 100, 300, 600, 1200$. Both graphical (Fig. 4.4) and numerical (Table 4.4) results indicate that when $\alpha = 0.3$ and $\beta = 0.03$, the extent of the epidemic is greater when the initial population size increases. Notably, the ratio $I_{\max} / S(0)$ increases to 0.96 when $S(0) = 1200$, indicating that 96% of the population would become infected by the disease.

85.0 Further Extensions to the Model: Mortality, Attenuation & Super-Spreaders

According to the World Health Organisation, Ebola results in a serious illness which is often fatal if left untreated. Official numerical data from [16] indicate that the average fatality rate is around 50%, and this value may vary between 25% to 90%. Logically, a further extension to the mathematical model would have to include a mortality parameter, μ , which indicates the proportion of infected individuals who do not recover from the disease.

It is proposed in this research that when mortality is taken into consideration, the original SIR model in Section 3.1 is modified to include the parameter μ , i.e.

$$\frac{dS(t)}{dt} = -\beta S(t)I(t) - \mu I(t) \quad (\text{Eq. 5.1})$$

$$\frac{dI(t)}{dt} = \beta S(t)I(t) - \alpha I(t) \quad (\text{Eq. 5.2})$$

$$\frac{dR(t)}{dt} = \alpha I(t) - \mu I(t) \quad (\text{Eq. 5.3})$$

Next, as a disease spreads through a given population, its level of virulence changes. According to [17], *attenuation* refers to the “change in the virulence of a pathogenic microorganism induced by passage through another host species, decreasing its virulence for the native host and increasing it for the new host.” Attenuation is the basis for the development of vaccines [18]. Since the strength of a pathogen varies with time, the transmission rate term, β , cannot be assumed to be constant but rather, a time-varying expression, $\beta(t)$, should be considered as part of the simulations. It is the subject of open debate as to whether a flu virus would weaken as it spreads through a given population but nevertheless, computer simulations could be utilised as a starting point to explore the potential effects of attenuation.

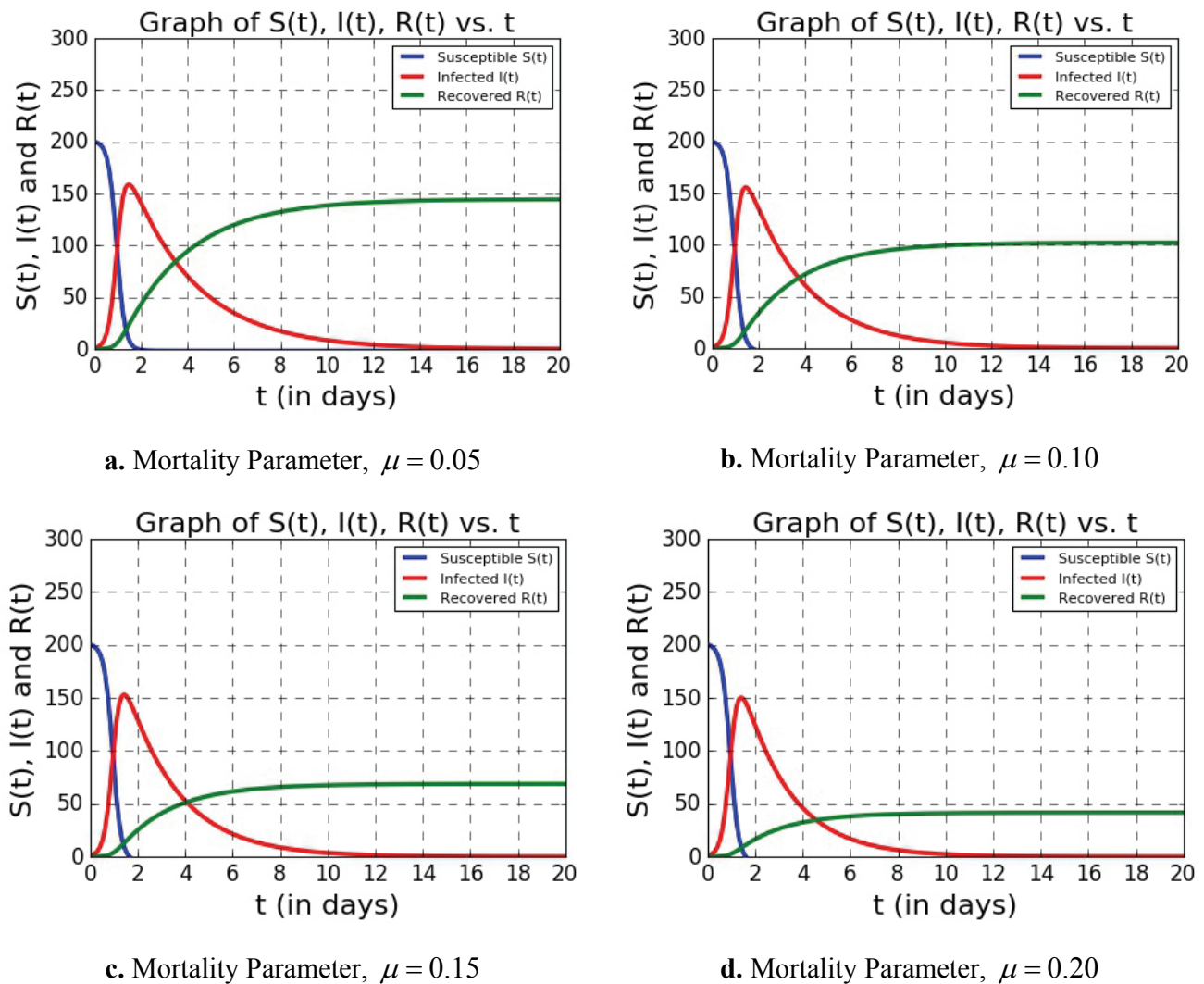
In our mathematical model, the original constant transmission rate term, β , is modified into a *time-varying* linear attenuation factor of the following form:

$$\beta(t) = \beta_0 - k \times t \quad (\text{Eq. 5.4})$$

where constant k determines the minimum time taken for the virulence of the disease to attenuate to zero, i.e. $\beta(t_{\min}) = 0$. Correspondingly, $t_{\min} = \frac{\beta_0}{k}$, where β_0 is the initial transmission rate.

Last but not least, the role of super-spreaders in epidemics should be taken into consideration. For instance, for SARS, more than 160 infections were linked to a single index case [12]. Therefore, a more accurate computer model would be one where the initial number of infected persons is larger than or much greater than unity, i.e. $I(0) \gg 1$.

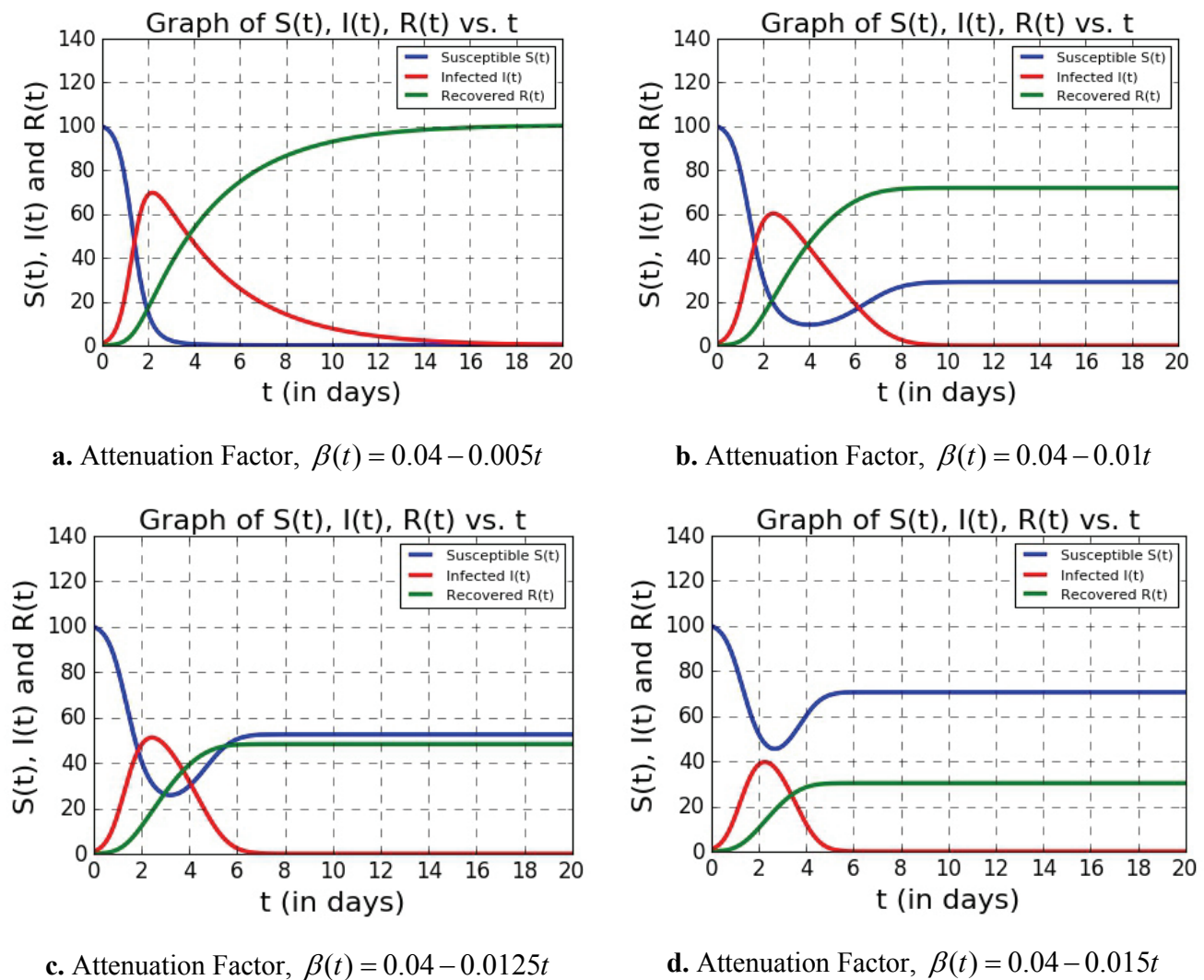
In this report, the additional scenarios of mortality, attenuation and super-spreaders will be examined in sub-sections 5.1, 5.2 and 5.3, respectively (c.f. Appendix V to VII).

5.1 Variation of Mortality Parameter μ

Figure 5.1: Simulation graphs for variation of mortality parameter μ .

$\mu (\text{day}^{-1})$	Initial Susceptible Persons $S(0)$	Max. Infected Persons $I_{\max}$ (persons)	Max. Recovered Persons $R_{\max}$ (persons)	Percentage of Max. Recovered Persons
0.05	200	159	145	73%
0.10	200	156	103	52%
0.15	200	153	69	35%
0.20	200	151	42	21%

Table 5.1: Numerical results for $R_{\max}$ with variation of mortality parameter μ .

Based on the simulation results in Fig. 5.1, the effects of mortality on overall population recovery is clear (i.e. $R_{\max} < I_{\max}$). Assuming the initial conditions $S(0) = 200$, $\alpha = 0.3$ and $\beta = 0.03$, mortality parameter μ is varied incrementally from 0.05 to 0.20 day^{-1} and numerical results are summarised in Table 5.1. As mortality increases, the maximum number of persons who recover from the disease decreases. From the simulation results, percentage of recovery falls below 50% when $\mu > 0.10 \text{ day}^{-1}$. At $\mu = 0.20 \text{ day}^{-1}$, percentage of recovery from infection is only 21%.

5.2 Linear Attenuation Factor $\beta(t)$

Figure 5.2: Simulation graphs for variation of linear attenuation factor $\beta(t)$.

Attenuation Factor	Min. Time for Full Attenuation $t_{\min}$ (days)	Max. Infected Persons $I_{\max}$ (persons)	Recovered Persons at $t = 20$ days $R(20)$	Susceptible Persons at $t = 20$ days $S(20)$
$\beta(t) = 0.04 - 0.005t$	8.00	70	100	0
$\beta(t) = 0.04 - 0.01t$	4.00	61	71	29
$\beta(t) = 0.04 - 0.0125t$	3.20	52	48	52
$\beta(t) = 0.04 - 0.015t$	2.67	40	30	70

Table 5.2: Numerical results for variation of linear attenuation factor $\beta(t)$.

In accordance with Eq. 5.4, the attenuation factor, $\beta(t)$, is varied and the minimum time taken for the virulent effects of the disease to attenuate to zero, is listed in Table 5.2. Examination of the numerical data indicate that as $t_{\min}$ decreases from 8.00 to 2.67 days, the maximum number of infected persons decreases and the number of susceptible (or uninfected) persons increase. These results indicate that when a disease attenuates in strength more quickly, its infectious effects decrease proportionally as well.

5.3 Super-Spreader Effect: Variation of Parameter $I(0)$

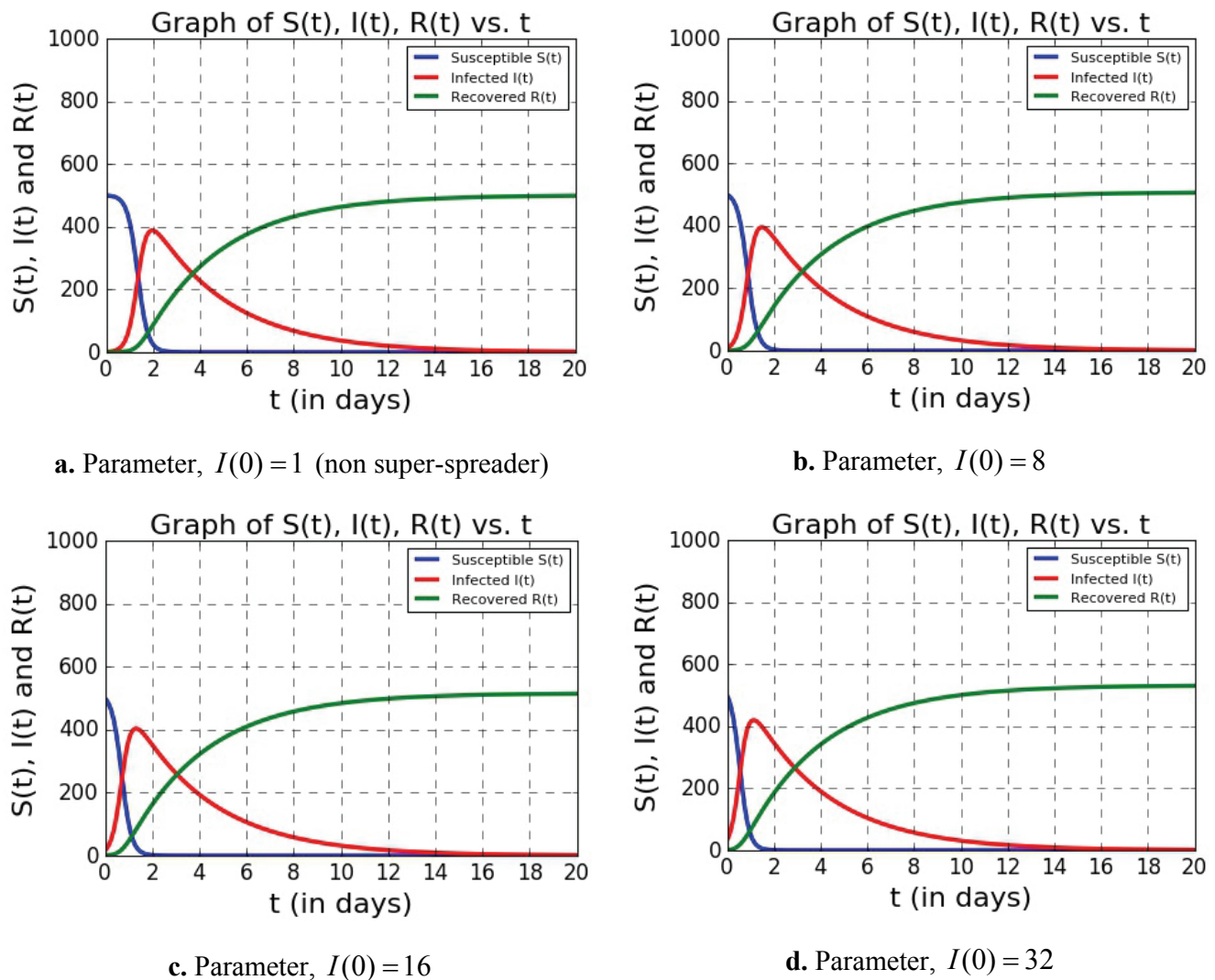


Figure 5.3: Simulation graphs for variation of initial infected population $I(0)$.

Initial Infected Persons $I(0)$	Max. Infected Persons $I_{\max}$ (persons)	Time to Max. Infection $t_{\max}$ (days)
1	389	1.97
8	396	1.50
16	404	1.33
32	420	1.14

Table 5.3: Numerical results for $I_{\max}$ and $t_{\max}$ with variation of $I(0)$.

To examine the effects of super-spreaders, the initial susceptible population has to be sufficiently large. Here, $S(0)$ is set to 500, the size of a hypothetical small town. Additionally, the case of $I(0) = 1$ (non super-spreader) has been included for reference. Assuming that $\alpha = 0.3$ and $\beta = 0.01$, the effect of super-spreaders (Figs. 5.3b–d) for a disease with average transmission rate is evident, i.e. the presence of a super-spreader that infects an initial number of 32 persons (viz. $I(0) = 32$) has reduced the time to maximum infection by nearly half, i.e. from approximately 2 days to 1.14 days (Table 5.3).

86.0 Summary of Key Results

In this report, a mathematical model of an epidemic based on the SIR model has been examined in detail, quantitatively, qualitatively and via computer simulations. The system of first-order differential equations in Eqs. 3.1–3.3 have been implemented algorithmically and relevant computer simulations carried out using Python 3.5 (c.f. Appendix I to IV) in order to obtain numerical results for determining the underlying trends of an epidemic.

To summarise, the following observations were made:

- $\frac{dS(t)}{dt}$ is proportional to the product of $S(t) \times I(t)$ because the expression can be understood based on the fact that the transmission of an infectious disease takes place via *direct contact* between a susceptible person and an infected one.
- $\frac{dR(t)}{dt}$ is a fraction of the number of infected persons, $I(t)$, where the constant of proportionality, α , determines the ratio of infected persons who subsequently recover from the disease after infection.
- α represents the fraction of infected persons who recover from the disease after being infected. It is otherwise known as the recovery parameter and has the units day^{-1} .
- β represents the transmission rate term, with virulent diseases having a higher numerical value for β . It has the units $\text{persons}^{-1} \text{ day}^{-1}$.
- For the special case where $\beta = 0$ and $\alpha \neq 0$, the number of susceptible (i.e. uninfected) persons remains constant and the number of infected persons decrease exponentially with time, in accordance with the derived equation $I = I_0 \exp(-\alpha t)$.
- An analytical solution for the recovered population is derived, i.e. $R(t) = I_0 \{1 - \exp(-\alpha t)\}$. A corollary of the expression is that $R = 0$ when $t = 0$, and $R \rightarrow I_0$ as $t \rightarrow \infty$.
- Based on Python 3.5 simulation results, when recovery parameter α increases, the maximum number of infected persons decreases. The stronger the immunity of a given population, the lesser the maximum number of persons will become infected by a particular disease.

- When β increases, the maximum number of infected persons increases rapidly, and the time taken to reach the point of maximum infection decreases. The transmission rate term β is greater for diseases that are more contagious.
- When $\alpha = 0.3$, $\beta = 0.03$, the extent of an epidemic is greater as the initial population size $S(0)$ increases.

With the above results in mind, the SIR model is further extended to include the effects of mortality, attenuation and super-spreaders (c.f. Appendix V to VII). For the first modification, the mortality parameter, μ , is added to Eqs. 5.1–5.3. With regard to attenuation, a linear attenuation factor of $\beta(t) = \beta_0 - k \times t$ is implemented in the computer simulations. Likewise, for the super-spreader effect, the initial number of infected persons, $I(0)$, is varied.

Based on these modifications, the following results were observed:

- As mortality increases, the maximum number of persons who recover from the disease decreases. Assuming the initial conditions $S(0) = 200$, $\alpha = 0.3$ and $\beta = 0.03$, percentage of recovery falls below 50% when $\mu > 0.10 \text{ day}^{-1}$.
- For a linear attenuation term, $\beta(t)$, when the time taken for full attenuation of the disease, $t_{\min}$, decreases from 8.00 to 2.67 days, the maximum number of infected persons decreases and the number of susceptible (or uninfected) persons increase. Results suggest that when a disease attenuates in strength more rapidly, its contagious effects decrease proportionally.
- Assuming $\alpha = 0.3$ and $\beta = 0.01$, the presence of a super-spreader that infects an initial number of 32 persons (viz. $I(0) = 32$) has reduced the time to maximum infection by nearly half, i.e. from approximately 2 days to 1.14 days.

These results indicate that the SIR model (Eqs. 3.1–3.3) is a robust mathematical model for analysing the spread of an epidemic. Furthermore, the model can be modified to take into account the effects of mortality, attenuation and super-spreaders (Eqs. 5.1–5.3). Additional scope for further extension of the model include other variations of attenuation term $\beta(t)$, e.g. quadratic, cubic, exponential etc. Also, the effects of vaccines on the spread of a disease could be studied computationally as well.

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Appendix I (Section 4.1)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 40
I_init = 30
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.0

# Euler's method

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k',linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, 50])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Appendix II (Section 4.2)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 100
I_init = 1
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.9
beta = 0.03

# Euler's method

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b') # plot P vs. time
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r') # plot G vs. time
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k',linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, 140])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Appendix III (Section 4.3)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 100
I_init = 1
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.12

# Euler's method

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k',linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, 140])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```


Appendix IV (Section 4.4)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 1200
I_init = 1
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.03

# Euler's method

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k',linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, S_init*1.5])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Appendix V (Section 5.1)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 200
I_init = 1
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.03
mu = 0.2

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I-mu*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I-mu*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k', linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, S_init*1.5])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

max_recovered = np.amax(X_arr[2,:])
print(max_recovered)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Appendix VI (Section 5.2)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 100
I_init = 1
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.04

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -(beta-0.015*t)*S*I
    dIdt = (beta-0.015*t)*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k', linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, S_init*1.4])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

print(X_arr[0,1280])

print(X_arr[2,1280])

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```

Appendix VII (Section 5.3)

```
import numpy as np
import matplotlib.pyplot as plt

print("Solution for Spread of Epidemic")

Dt = 1/64
S_init = 500
I_init = 32
R_init = 0
t_init = 0
t_end = 20

n_steps = int(round((t_end-t_init)/Dt))

X = np.zeros(3)
dXdt = np.zeros(3)
t_arr = np.zeros(n_steps+1)
X_arr = np.zeros((3,n_steps+1))
t_arr[0] = t_init
X_arr[0,0] = S_init
X_arr[1,0] = I_init
X_arr[2,0] = R_init

alpha = 0.3
beta = 0.01

for i in range(1,n_steps+1):
    t = t_arr[i-1]
    S = X_arr[0,i-1]
    I = X_arr[1,i-1]
    R = X_arr[2,i-1]
    X[0] = S
    X[1] = I
    X[2] = R
    dSdt = -beta*S*I
    dIdt = beta*S*I-alpha*I
    dRdt = alpha*I
    dXdt[0] = dSdt
    dXdt[1] = dIdt
    dXdt[2] = dRdt
    Xnew = X + Dt*dXdt
    X_arr[:,i] = Xnew
    t_arr[i] = t + Dt

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 3, label="Susceptible S(t)", color='b')
plt.plot(t_arr, X_arr[1,:], linewidth = 3, label="Infected I(t)", color='r')
plt.plot(t_arr, X_arr[2,:], linewidth = 3, label="Recovered R(t)", color='g')
plt.xticks(np.arange(0, t_end+1, 2.0))
plt.grid(b=True, which='both', color='k',linestyle='--')

plt.title('Graph of S(t), I(t), R(t) vs. t', fontsize = 20)
plt.xlabel('t (in days)', fontsize = 20)
plt.ylabel('S(t), I(t) and R(t)', fontsize = 20)

plt.xticks(fontsize = 15)
plt.yticks(fontsize = 15)
plt.axis([0, t_end, 0, S_init*2])

plt.legend(fontsize=10)
plt.show()

max_infected = np.amax(X_arr[1,:])
print(max_infected)

np_array = np.array(X_arr[1,:])
item_index = np.where(np_array==max_infected)
print(item_index)

fig.savefig('epidemic.jpg', dpi=fig.dpi, bbox_inches = "tight")
```